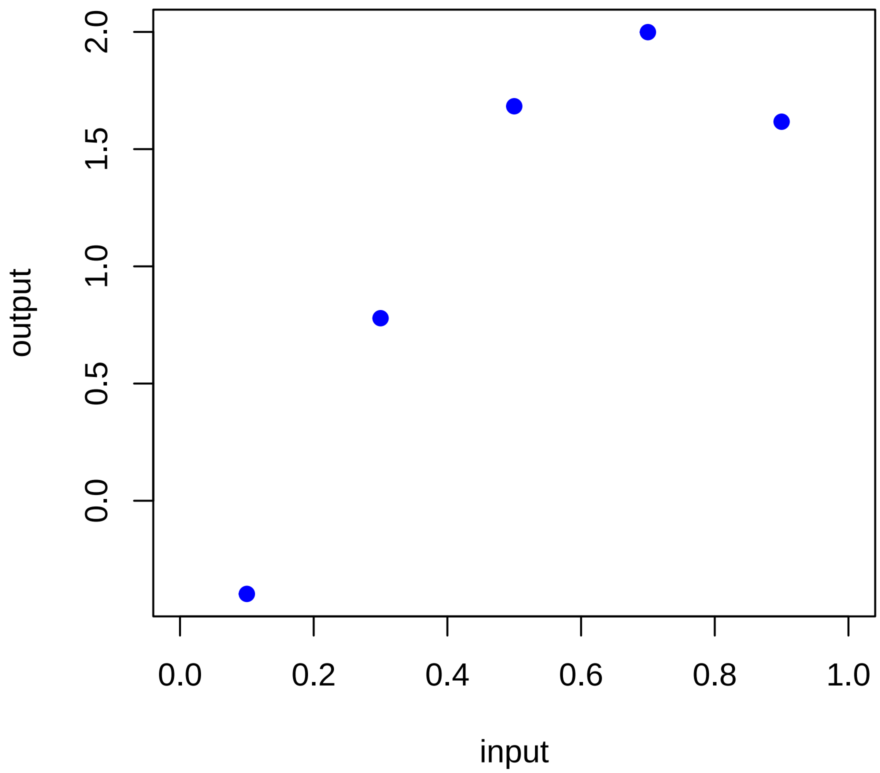
RobustGaSP 1D example

We have a 1-dimensional input, one output, and 5 noise-free observations, obtained using the function $f(x) = 2\sin(3x - 0.5)$:

$$x = \{0.1, 0.3, 0.5, 0.7, 0.9\}$$

$$y = \{-0.4, 0.78, 1.68, 2, 1.62\}$$

which, when plotted, look like this:



We fit an emulator using the `rgasp()` function in the `RobustGaSP` package:

```
rgasp(design = input,
      response = output,
      trend = H_matrix,
      lower_bound=FALSE,
      kernel_type = 'pow_exp', alpha = 2,
      zero.mean = "No"
      )
```

Here, a constant "mean basis" has been used, with "no constraint on optimization" (as per the `RobustGaSP` package manual). A Gaussian (squared exponential) kernel has been used, and linear mean prior function.

Should 'zero.mean' equal "No"?

We get the following output:

```
## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization  1  :
## The number of iterations is  44
##  The value of the  marginal posterior  function is  -0.2280412
##  Optimized range parameters are 3.655377
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 0.064
## Start of the optimization  2  :
## The number of iterations is  25
##  The value of the  marginal posterior  function is  4.466659
##  Optimized range parameters are 1.114853
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
```

Contrast this with using a Matern 5/2 instead:

```
## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization  1  :
## The number of iterations is  21
##  The value of the  marginal posterior  function is  -5.227341
##  Optimized range parameters are 619828629467
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 0.064
## Start of the optimization  2  :
## The number of iterations is  16
##  The value of the  marginal posterior  function is  3.12738
##  Optimized range parameters are 4.677144
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
```

What are the two sets of optimisations here? Is it a problem that we sometimes get `TRUE` for both, sometimes only for one?

The posterior mean function for both kernels, as well as that obtained using both kernels in `DiceKriging`'s `km()` function, again with a linear mean prior function, are shown in Figure 1.

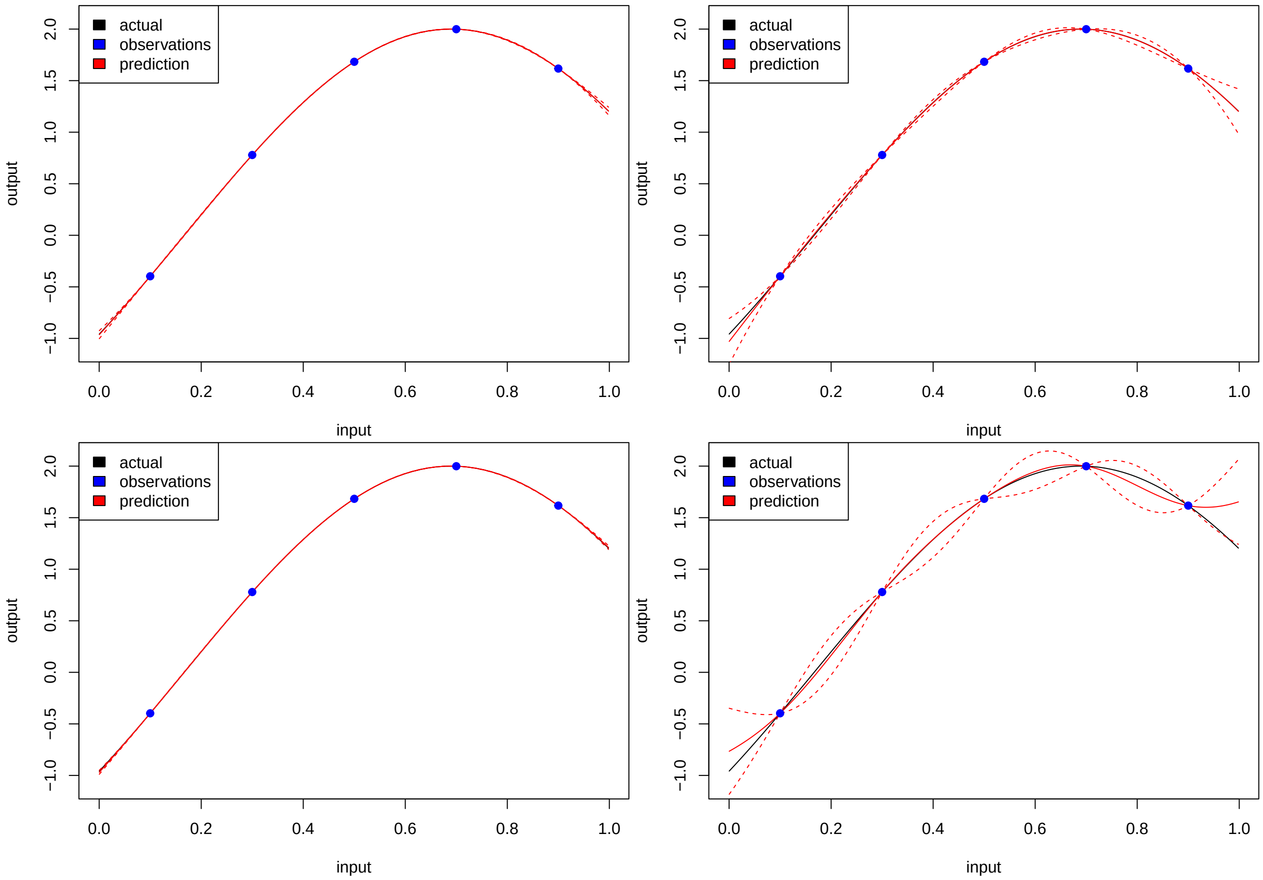


Figure 1: Top: RobustGaSP with Gauss kernel (left) and Matern 5/2 (right). Bottom: DiceKriging with Gauss kernel (left) and Matern 5/2 (right).

Various checks are performed (as in the example from `RobustGaSP`'s Help page):

Package	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
RobustGaSP	Gauss	1	0.0000009	0.0103
RobustGaSP	Matern 5/2	1	0.0002	0.0822
DiceKriging	Gauss	1	0.000005	0.0052
DiceKriging	Matern 5/2	0.97	0.008	0.2696

Finally, the estimates returned by each of these four setups are shown below. For `RobustGaSP`, the Manual indicates that it returns "inverse-range parameters", so the inverse of these values have been found, resulting in ℓ_{RK} (first table below).

In the Gauss kernel case, the Manual for `RobustGaSP` indicates that the denominator of the fraction is ℓ_{RK}^2 whereas for `DiceKriging` it's $2\ell_{DK}^2$. To facilitate direct comparison of the parameter estimates, we require a range parameter for `RobustGaSP`, ℓ which, when plugged into $2\ell^2$ results in ℓ_{RG}^2. Therefore

$$\ell = \frac{\ell_{RG}}{\sqrt{2}},$$

since

$$2\left(\frac{\ell_{RG}}{\sqrt{2}}\right)^2 = 2\frac{\ell_{RG}}{2}^2 = \ell_{RG}^2,$$

shown in the second table below.

Package	Kernel	Trend 1	Trend 2	Range	Variance
RobustGaSP	Gauss	-0.7	-0.57	0.79	15.1
RobustGaSP	Matern 5/2	-19.51	-2.65	4.68	2754.05
DiceKriging	Gauss	-1.13	0.18	0.69	5.05
DiceKriging	Matern 5/2	-0.37	2.43	0.28	0.35

Package	Kernel	Trend 1	Trend 2	Range	Variance
RobustGaSP	Gauss	-0.7	-0.57	1.11	15.1
RobustGaSP	Matern 5/2	-19.51	-2.65	4.68	2754.05
DiceKriging	Gauss	-1.13	0.18	0.69	5.05
DiceKriging	Matern 5/2	-0.37	2.43	0.28	0.35

Are the range of values here worrying? Have I done everything correctly here?

Verifying the estimates by-hand for both packages

We already have matrix H (defined earlier for use in the `rgasp` function's `trend` argument):

```
##      [,1] [,2]
## [1,]    1  0.1
## [2,]    1  0.3
## [3,]    1  0.5
## [4,]    1  0.7
## [5,]    1  0.9
```

We also need matrix A and its inverse, built using the estimate of the range parameter.

Finally, we need $\mathbf{t}(x^*)$ for all 1000 of the new inputs x^*.

We're now ready to calculate posterior predictions for each of the new input settings, using the posterior estimates for the trend parameters (returned by `DiceKriging` using the Gauss kernel). By-hand predictions match those returned by `DiceKriging` (Figure 2).

This is repeated for `RobustGaSP` with the Gaussian kernel. The Manual says that it returns the *inverse* of the range parameters, so the ones returned here have had their inverses found. The Manual also indicates that there is not a 2 in the denominator of the Gauss kernel function, so this has also been removed. Again, by-hand predictions match those returned by `DiceKriging` (Figure 3).

For both we calculate the following:

$$\frac{\Sigma_{i=1}^{1000} \left|(\text{predictions from package})_i - (\text{predictions by-hand})_i\right|}{10^{-8} \times 1000}$$

and get, for `DiceKriging` and `RobustGaSP` respectively:

```
## [1] 0.0006134877
```

```
## [1] 0.001868237
```

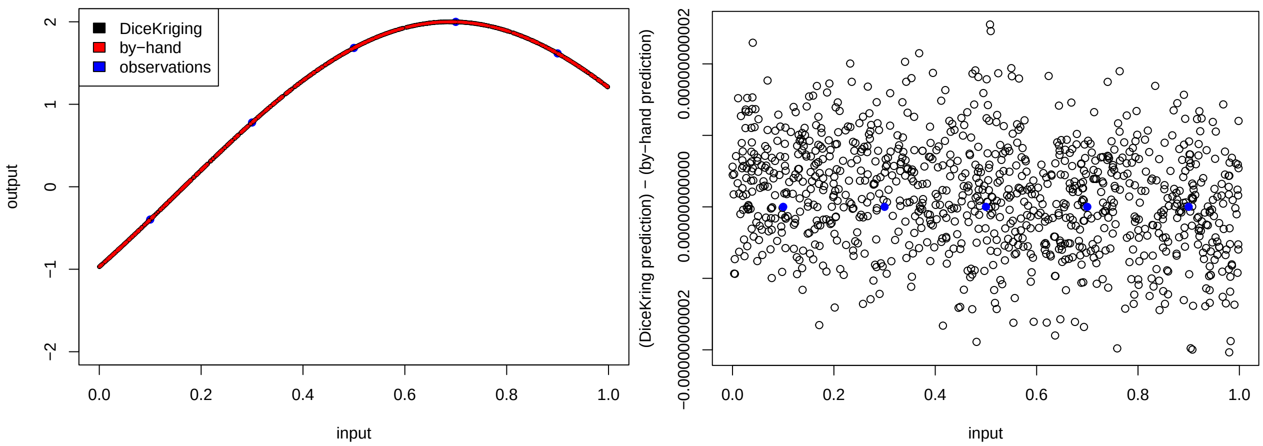


Figure 2: The predictions found by-hand match well with those made by DiceKriging (left), with the latter minus the former found to be essentially zero (right).

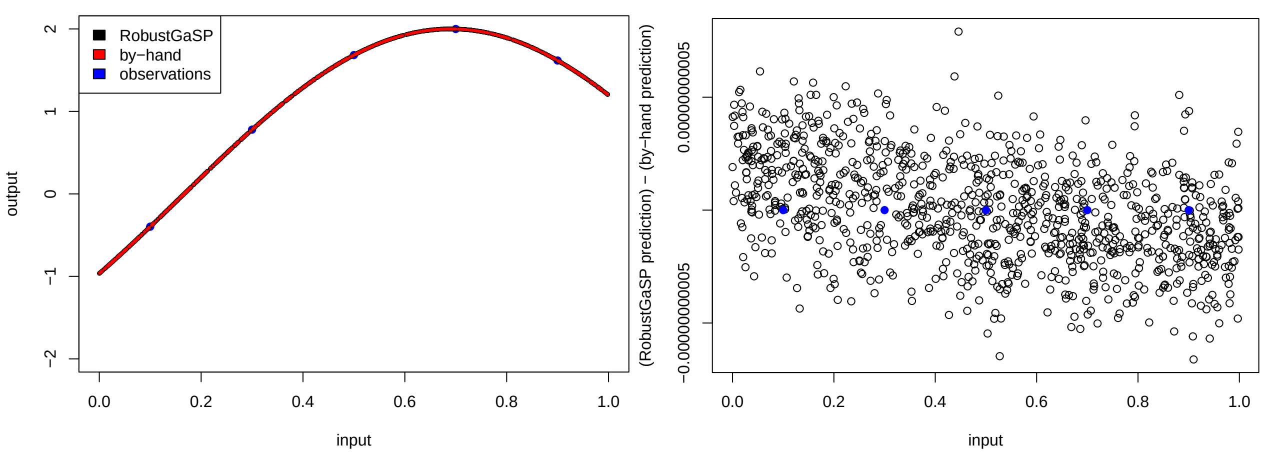


Figure 3: The predictions found by-hand match well with those made by RobustGaSP (left), with the latter minus the former found to be essentially zero (right).